

Atomic Structure & the Periodic Table

Test · 36 marks total

Name: _____

Date: _____

Score: _____ / 36

Section A — Atoms, particles & isotopes

A1. State the relative charge of a proton, and the relative charge of an electron.

[2 marks]

A2. An atom of an element has 6 protons and 6 neutrons. State its atomic number and its mass number.

[2 marks]

A3. An atom of chlorine has atomic number 17 and mass number 35. Calculate its number of neutrons.

[1 mark]

A4. Explain what is meant by the term “isotopes”.

[2 marks]

A5. Bromine has two isotopes: bromine-79 (50.7% abundance) and bromine-81 (49.3% abundance). Calculate the relative atomic mass of bromine, to 1 decimal place.

[3 marks]

Section B — Electronic structure

B1. State the maximum number of electrons that can occupy the 1st shell, and the 2nd shell.

[2 marks]

B2. Give the electronic structure of a phosphorus atom (atomic number 15).

[2 marks]

B3. An atom has the electronic structure 2,8,7. State which group and which period this atom belongs to.

[2 marks]

Section C — Development of the periodic table

C1. Describe Newlands’ “Law of Octaves”, and give one reason it was not widely accepted.

[2 marks]

C2. Explain how Mendeleev used gaps in his periodic table to gain scientific credibility for his arrangement of the elements.

[3 marks]

C3. State the property used to arrange elements in the modern periodic table.

[1 mark]

C4. State what elements in the same group of the periodic table have in common, in terms of electronic structure.

[1 mark]

Section D — Metals, non-metals & key groups

D1. Give two physical properties of metals.

[2 marks]

D2. Give two physical properties of non-metals.

[2 marks]

D3. Explain why the noble gases (Group 0) are very unreactive.

[2 marks]

D4. Sodium reacts with water. (a) Write a word equation for this reaction. (b) Describe two observations you would make during the reaction.

[3 marks]

D5. Explain, in terms of electronic structure, why reactivity increases down Group 1.

[2 marks]

D6. Chlorine can displace bromine from a solution of potassium bromide. Write a word equation for this reaction, and state what type of reaction this is.

[2 marks]